

Rainfall Monitoring System – MVP Design Document

Project Goal

Build a mobile application prototype for rainfall monitoring that allows field operators to submit rainfall data from designated locations and enables district/state officers to monitor reporting status through GIS-based dashboards.

The MVP will focus on one district (Raipur) with a limited number of stations while keeping the architecture scalable for future deployment across all rainfall stations in Chhattisgarh.

MVP Scope

Geographic Scope

State: Chhattisgarh

District: Raipur

Administrative Hierarchy:

State

└── District

└── Block

└── Village

└── Rainfall Station

MVP Dataset

- 1 District (Raipur)
 - 3 Blocks
 - 8–10 Villages
 - 10 Rainfall Stations
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Core Objectives

1. Manual rainfall data entry.
2. GPS-based station verification.

3. Role-based hierarchical access.
 4. Offline data storage.
 5. GIS visualization.
 6. Missing station monitoring.
 7. Dashboard aggregation.
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User Roles

1. Field Operator

Responsibilities:

- Submit rainfall data.
- View assigned station.
- View own submission history.

Permissions:

- Cannot view other stations.
 - Cannot access district/state dashboards.
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2. Block Officer

Assigned to:

- One block

Responsibilities:

- Monitor all villages within the block
- Track reporting status
- Identify missing stations
- View rainfall summaries

Permissions:

- ✓ View all villages in assigned block
- ✓ View all stations in assigned block
- ✓ View block rainfall analytics
- ✓ Monitor missing reports

✗ Cannot access other blocks

✗ Cannot access state-wide data

Example:

Abhanpur Block

|—— Village A
|—— Village B
|—— Village C
|—— Village D

Dashboard:

Block: Abhanpur

Villages: 4

Stations: 10

Reported: 8

Missing: 2

Average Rainfall: 18 mm

3. District Officer

Responsibilities:

- Monitor all stations within Raipur district.
- Identify missing stations.
- View rainfall summaries.

Permissions:

- View all blocks.
 - View all villages.
 - View all stations within district.
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4. State Admin

Responsibilities:

- Monitor district-level reporting.
- View overall reporting status.
- Access GIS dashboard.

Permissions:

- Full visibility.
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Authentication

Login Options

1. Mobile Number + OTP
2. Username + Password

MVP Implementation

Use mock authentication:

Field Operator

operator_rp001 / 123456

District Officer

district_raipur / 123456

State Admin

state_admin / 123456

Device Binding (Future Production Feature)

Each account should be linked to one device.

MVP:

- Generate device ID.
 - Display device information.
 - Simulate device registration.
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GPS Verification

Requirement

User must be physically present near the assigned rainfall station.

MVP Rule

Allowed Radius:
20 meters

Workflow:

Select Station
↓
Get Current GPS
↓
Calculate Distance
↓
Distance $\leq$ 20m
↓
Allow Submission

Otherwise:
Show warning and disable submission.

Rainfall Submission

Fields

- Rainfall (mm)
- Timestamp
- Latitude
- Longitude
- Station ID
- Remarks (optional)

Example

Station: RP001
Rainfall: 22 mm
Time: 09:30 AM

Offline Support

MVP

Store entries locally.

Status:

- Synced
- Pending Sync

Workflow:

Submit

↓

Save Locally

↓

Internet Available

↓

Sync

Station Status System

Each station should have one of the following states:

Green

Reported Successfully

Meaning:

Rainfall submitted today.

Red

Missing Data

Meaning:

No rainfall entry received today.

Yellow

Pending Sync

Meaning:

Data entered but not uploaded yet.

GIS Dashboard Concept

This is the primary feature of the system.

State Dashboard

Display a Chhattisgarh map.

Each district is represented with color coding.

Example:

Green = Data Available

Yellow = Partial Reporting

Red = Missing Reporting

User can click a district.

District Dashboard

Display Raipur district map.

Show all rainfall stations as markers.

Marker Colors:

Green

Submitted

Red

Missing

Yellow
Pending Sync

Example:

RP001 → Green
RP002 → Green
RP003 → Red
RP004 → Yellow

Dashboard Metrics

District Dashboard

Total Stations

Reported Today

Missing Stations

Pending Sync

Average Rainfall

Highest Rainfall Station

Data Aggregation

Rainfall data flows upward.

Station
↓
Village
↓
Block
↓
District
↓
State

Dashboard summaries should be generated from station-level data.

MVP Screens

1. Splash Screen
2. Login Screen
3. Home Dashboard
4. Station List
5. Station Detail Screen
6. Map & GPS Verification Screen
7. Rainfall Entry Screen
8. Offline Queue Screen
9. District GIS Dashboard
10. State GIS Dashboard
11. Profile Screen

Flutter Technology Stack

Framework:
Flutter

State Management:
Riverpod

Navigation:
Go Router

Maps:
Flutter Map

Location:
Geolocator

Storage:

Hive

Charts:

fl_chart

Permissions:

permission_handler

MVP Demo Flow

Login



Open Assigned Station



Map Displays Station



GPS Validation



Submit Rainfall



Station Marker Turns Green



District Dashboard Updates

If no submission exists:

Station Marker Remains Red

This demonstrates rainfall collection, GPS verification, hierarchical monitoring, and GIS-based reporting.

Future Production Enhancements

- Full Chhattisgarh deployment (734 stations)
- Backend APIs
- PostgreSQL + PostGIS
- Real-time updates
- Device binding
- Push notifications
- Missing data alerts
- Heatmaps
- Analytics dashboards

- Role management
- Audit logs
- Offline synchronization engine